Marketing Analytics Project
README Documentation

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1 Overview

This project provides a comprehensive analysis pipeline for Marketing Analytics. The script covers multiple tasks including:

- Data exploration and preparation
- RFM modeling
- Churn prediction modeling
- Sentiment analysis

The analysis is organized by data tables, with dedicated sections for each dataset. This modular approach ensures clarity, scalability, and reproducibility.

2 Libraries Used

This section lists the Python libraries used across the analysis, modeling and visualization steps. They are grouped by purpose to make environment setup and dependency management easier.

Core data and utilities

- **pandas** — data manipulation, merging, grouping, time handling
- **numpy** — numerical operations and array handling
- **datetime** — date/time utilities (also using `pandas.Timestamp`)
- **joblib** — model and object serialization
- **ast**, **json**, **difflib**, **re**, **string** — parsing, fuzzy matching, text cleaning

Visualization

- **matplotlib** (`pyplot`) — plotting (histograms, line charts, bar charts)
- **seaborn** — higher-level statistical visualizations and styles

Machine learning & modeling

- **scikit-learn** — train/test split, preprocessing (LabelEncoder/OrdinalEncoder), metrics, classical models (Logistic Regression, RandomForest), TF-IDF
- **lightgbm** — gradient boosting models for churn prediction
- **imblearn (SMOTE)** — synthetic oversampling to handle class imbalance
- **joblib** — model persistence

Text / NLP

- **nltk** (stopwords, WordNetLemmatizer) — tokenization, stopword removal, lemmatization
- **tensorflow.keras** (Tokenizer, pad_sequences, Sequential, Embedding, Conv1D, GlobalMaxPooling1D, Dense, Dropout) — deep learning models for text (CNN-based)
- **sklearn.feature_extraction.text.TfidfVectorizer** — TF-IDF features for classical text models
- **sklearn.naive_bayes.MultinomialNB** — baseline text classifier

Evaluation & utilities

- **scikit-learn.metrics** — ROC AUC, Precision-Recall, classification reports, confusion matrices, plotting utilities (RocCurveDisplay, PrecisionRecallDisplay)
- **matplotlib / seaborn** — plotting model diagnostics and feature importances

Suggested installation (example)

You can install the main packages with `pip` (adjust versions as needed):

```
pip install pandas numpy matplotlib seaborn scikit-learn lightgbm imbalanced-learn jo
```

Note: depending on your environment, you may prefer `conda` for TensorFlow and LightGBM installations. Also remember to download NLTK resources (e.g., `stopwords`, `wordnet`) if you use NLTK:

```
import nltk
nltk.download('stopwords')
nltk.download('wordnet')
```

3 Users Analysis

This section is based on the `USERS` table. It prepares and explores the data, focusing on customer demographics, behaviors, and key indicators.

Contents

- Data preparation
- User growth trend
- Customer acquisition channels
- Anonymous vs Registered users
- Geography and Language
- Registration modality
- KPI

4 Profiles Analysis

Exploration of user profiles to understand demographics, physical characteristics, ski experience, service preferences, and key engagement metrics.

- Data Preparation
- Age Distribution
- Sex Distribution
- City Distribution
- Height and Weight Analysis
- Ski Level
- Types of Services
- KPI

5 Cards Analysis

This section is based on the `CARDS` table. It investigates customer profiles and card-related behaviors.

Contents

- Data preparation
- Card status
- Demographic analysis
- Buyers for themselves vs buyers for others
- Combined analysis: age $\times$ status
- KPI

6 Orders Analysis

This section is based on the `ORDERS` table. It focuses on order-level data, logistics, and user purchasing behaviors.

Contents

- Data preparation
- Order trends over time
- Purchase channels (App vs Web)
- Payment methods
- Pickup and logistics
- Delay in payment
- User segmentation
- KPI

7 Order Details Analysis

This section is based on the `ORDER_DETAILS` table. It investigates product-level insights and purchasing behaviors.

Contents

- Data preparation
- Most popular products and areas
- Seasonality of products
- Cancellation rate and booking behavior
- Typical duration of experiences
- Co-occurrence / Market Basket Analysis (simple heatmap)
- Basket size
- Seasonality by product type
- KPI

8 Data Understanding

This section focuses on gaining a deeper understanding of the relationships between different tables (`USERS`, `ORDERS`, `ORDER_DETAILS`, `PROFILES`, and `CARDS`). The objective is to validate data consistency, summarize user behaviors, and compute season-based metrics.

Contents

- Unique users who placed at least one order vs. total registered users
- Orders with detailed items in the `ORDER_DETAILS` table
- Number of profiles and cards per user (summary statistics)
- Consistency checks:
  - Users in `ORDERS` that also exist in `USERS`
  - Orders with missing user IDs in `USERS`
- Season-based analysis (2023–2024 season: Nov 1, 2023 to Apr 30, 2024):
  - Average Order Value (AOV)
  - Total spending per user
  - Average spending per user
- User activity by season
- Churn analysis between consecutive seasons (e.g., 2023/24 → 2024/25)

9 RFM Analysis

This section applies an **RFM (Recency, Frequency, Monetary)** model to segment customers based on their purchasing behavior. The analysis uses valid orders (with status: `ok`, `fulfilled`, or `on-hold`) and computes RFM scores and segments.

Contents

- Data preparation:
  - Filtering valid orders and aggregating order amounts
  - Merging orders with user information
  - Ensuring proper date formatting
- Construction of the RFM table:
  - **Recency**: days since last order
  - **Frequency**: number of unique orders
  - **Monetary**: total spending
- RFM scoring:
  - Quartile-based scores for each dimension (1–4 scale)
  - Computation of composite scores (`RFM_code`, `RFM_sum`)
- Customer segmentation rules:
  - Champions

  - Loyal
  - Big Spenders
  - Frequent Buyers
  - New / Promising
  - At Risk
  - Lost
  - Low Value / Occasional

- Outputs and visualizations:
  - Distribution of Recency (histogram)
  - Customers by segment (bar chart)
  - Top 10 customers by Monetary value
  - Segment-level metrics: % Customers vs. % Revenue (side-by-side bar chart)

- Export of RFM results to `rfm_customers_quartiles.csv`

10 Churn Prediction Model

This section develops a machine learning pipeline to predict customer churn, combining feature engineering, churn labeling, and model training. The approach leverages transactional, behavioral, and demographic data across multiple tables (`USERS`, `ORDERS`, `ORDER_DETAILS`, `PROFILES`, `CARDS`).

Contents

- **Data Preparation**
  - Cleaning and aligning key identifiers across tables
  - Standardizing timestamps and deriving reference dates (season-based)
  - Filtering valid orders and enriching with order details
  - Feature engineering:
    * Purchase behavior (order counts, recency, tenure, orders per month)
    * Monetary metrics (total spend, average spend, discounts, dynamic pricing)
    * Product engagement (unique products, zones, duration of experiences)
    * Demographics (age, gender ratio, number of profiles)
    * Cards usage (count and status distribution)

- **Churn Labeling**
  - Seasons defined as November–April
  - A user is labeled as `churned` if active in season t but inactive in season $t + 1$
  - Churn label added at user level (`churn` = 0 retained, 1 churned)

- **Exploratory Churn Analysis**

  - Seasonal active users
  - Retention vs. churn visualization (stacked bars)
  - Computation of churn rates across seasons

- **Modeling Approaches**
  - **Baseline Model:** LightGBM with class weighting
  - **Imbalance Handling:**
    * Downsampling of majority class
    * SMOTE oversampling with categorical encoding
  - **Model Comparison:**
    * LightGBM
    * Random Forest
    * Logistic Regression

- **Evaluation**
  - Metrics: ROC AUC, PR AUC, classification report, confusion matrix
  - Visualizations: ROC curve, Precision–Recall curve
  - Feature importance ranking (top 30 predictors)
  - Comparative bar chart of ROC AUC vs. PR AUC across models

11 Sentiment Analysis

The last part of the project focuses on analyzing customer reviews:

- **Data Preparation:** loading datasets (`reviews.csv`, `reviews_labelled.csv`), null value checks, label distribution
- **Text Preprocessing:** tokenization, stopword removal, lemmatization, punctuation and number filtering
- **Feature Engineering:** TF-IDF vectorization and padded sequences for deep learning
- **Models:**
  - Logistic Regression
  - Multinomial Naive Bayes
  - Convolutional Neural Network (CNN with embeddings, Conv1D, pooling, dense layers)
- **Evaluation:** accuracy, classification report, confusion matrices
- **Application:** prediction of sentiments on unlabelled reviews, storage of results, user-level sentiment distribution
- **Visualization:** bar plots, pie charts, most negative/positive users, longest reviews